KAPASA MAKASA UNIVERSITY

FACULTY OF APPLIED SCIENCE AND OPEN LEARNING

DEPARTMENT OF INFORMATION & COMMUNICATION TECHNOLOGY

ISB 140/DS 110 – INTRODUCTION TO STATISTICS

Tutorial sheet 1

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1. A sample of 25 measurements is given here:

   9 28 14 29 21 27 15 23 23 10 31 23 16 26 22

   17 19 24 21 20 26 20 16 14 21.

   a) Make a frequency table displaying class intervals, frequencies, relative frequencies, and percentages.

   b) Construct a frequency histogram and interpret.

2. Prepare a box-and-whisker plot for the following data:

   36 43 28 52 41 59 47 61 24 55 63 73 32

   25 35 49 31 22 61 42 58 65 98 34

   Does this data set contain any outliers?

3. A survey of families living in a certain city was conducted to collect information on the following variables: age of the oldest person in the family, number of family members, number of males in the family, number of females in the family, whether or not they own a house, income of the family, whether or not the family took vacations during the past one year, whether or not they are happy with their financial situation, and the amount of their monthly mortgage or rent.

   a) Which of these variables are qualitative variables?

   b) Which of these variables are quantitative variables?

   c) Which of the quantitative variables of part (b) are discrete variables?

   d) Which of the quantitative variables of part (b) are continuous variables?

4. The following data give the mean of SAT Mathematics scores for a randomly selected 20 states.

   558 503 565 572 546 517 542

   605 493 499 568 553 510 525

   595 502 526 475 506 568. Construct

   a) A stem-and-leaf display and interpret.

   b) A frequency histogram and interpret.

   c) A pie chart and interpret.

5. According to the recent U.S. Federal Highway Administration *Highway Statistics*, the percentages of freeways and expressways in various road mileage–related highway pavement conditions are as follows: Poor 10%, Mediocre 32%, Fair 22%, Good 21%, and Very good 15%. Construct

   a) a bar graph.

   b) a pie chart.

6. Indicate which of the following variables are quantitative and which are qualitative.

   a) The amount of time a student spent studying for an exam

   b) The amount of rain last year in 30 cities.

   c) The arrival status of an airline flight (early, on time, late, cancelled) at an airport

   d) A person's blood type.

   e) The amount of gasoline put into a car at a gas station.

7. Construct a stem and leaf plot for these 50 measurements

   3.1 4.9 2.8 3.6 2.5 4.5 3.5 3.7 4.1 4.9 2.9 2.1 3.5 4.0 3.7 2.7 4.0 4.4 3.7 4.2

   3.8 6.2 2.5 2.9 2.8 5.1 1.8 5.6 2.2 3.4 2.5 3.6 5.1 4.8 1.6 3.6 6.1 4.7 3.9 3.9

   4.3 5.7 3.7 4.6 4.0 5.6 4.9 4.2 3.1 3.9

   a) Describe the shape of the data distribution. Do you see any outliers?

   b) Use the stem and leaf plot to find the smallest observation.

   c) Find the eighth and ninth largest observations.

8. The distance travelled (in kilometres) by a courier service motorcycle on 30 trips is recorded by the driver

24	19	21	27	20	17	17	32	22	26
18	13	23	30	10	13	18	22	34	16
18	23	15	19	28	25	25	20	17	15

   Construct

   a) A histogram.

   b) A polygon

c) An ogive.

d) The relative ogive

9. The table below gives the frequency distribution of the number of orders received each day during the past 50 days at the office of a mail-order company.

Number of orders	**Number of days (f)**
10 – 12	4
13 – 15	12
16 – 18	20
19 – 21	14

Estimate

a) The lower and upper quartiles,

b) The standard deviation.

10. A class of students played a computer game which tested how quickly they reacted to a visual instruction to press a particular key. The computer measured their reaction times in tens of a second and stored a record of a sex and reaction time of each student. Finally, it displayed the following summary statistics for the whole class.

Sex	Median	Lower quartile	Upper quartile	Min	Max
Female students	10	8	15	6	19
Male students	10	7	13	4	16

a) Draw two box plots suitable for comparing the reaction times of male and female students.

b) Write a brief comparison of the performance of male students and female students in this game.

11. The following data give the letter grades of 20 students enrolled in a statistics course

A	B	F	A	C	C	D	A	B	F
C	D	B	A	B	A	F	B	C	A

a) Construct a bar graph.

b) Construct a pie chart.

12. For each of the following ungrouped data, find the mean, mode and Median of the following dataset.

a) 40, 46, 40, 54, 18, 45, 34, 60, 39, 42

b) 52, 61, 78, 49, 47, 79, 54, 58, 62, 73, 72

c) 4,6, 18, 25, 9, 16, 22, 5, 20, 4, 8

13. The report "Motorcycle Helmet included observations made on the helmet use of 1700 motorcyclists across the country. The data is summarized in the accompanying table.

Category	Frequency
No helmet	731
No compliant helmet	153
Compliant helmet	816
Total	1700

a) What is the variable in this example? What are the possible values of that variable?

b) Construct the relative frequency distribution of this variable.

c) What proportion of observed motorcyclists wore some kind of helmet?

d) Construct an appropriate graph for this data.

14. A certain article reported on a pilot study in which each individual in a sample was asked to estimate the calorie, and the following information appeared in the article:

Class	Percentage
$0-50$	7
$50-75$	9
$75-100$	23
$100-125$	31
$125-150$	12
$150-200$	3
$200-300$	12
$300-500$	3

a) Construct a histogram of the data and comment on any interesting features.

b) What proportion of the estimates were at least 100? Less than 200?

15. Consider the following data below

74 52 67 68 71 76 86 81 73 68 64 75 71 61 63 57 67 57 59 72 79 64 70 74 77 79 65 68 69 83

a) Draw a stem plot to show the masses, correct to the nearest kilograms of 30 men use intervals $50-54,\ 55-59,\ 60-64, ...$

b) Write down the modal mass and calculate the mode

c) Compute the mean and median.

16. The pulse rate of 30 company directors were measured before and after ranking exercise, and the pulse rate data is displayed as follows:

Before: 110, 93, 81, 75, 73, 73, 48, 53, 69, 69, 66, 111, 105, 93, 90, 50, 57, 64, 90, 111, 91, 70, 70, 51, 79, 93, 105, 51, 66, 93

After: 117, 81, 77, 108, 130, 69, 77, 84, 84, 86, 95, 125, 96, 104, 104, 137, 143, 70, 80, 131, 145, 106, 130, 109, 137, 75, 104, 75, 97, 80.

(Use class intervals $40-49, 50-59, 60-69, ...$).

Draw a back-to-back leaf and stem plot for the given data above.

17. Identify each variable as nominal, ordinal, interval and ratio scales

a) UK political party preference (Labour, Conservative, Social Democrat).

b) Anxiety rating (none, mild, moderate, severe, very severe).

c) Patient survival (in number of months).

d) Clinic location (London, Boston, Madison, Rochester, Montreal).

e) Response of tumour to chemotherapy (complete elimination, partial reduction, stable, growth progression).

f) Favourite beverage (water, juice, milk, soft drink, beer, wine).

g) Appraisal of company's inventory level (too low, about right, too high).